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The analysis—Hierarchical models: Past, present and future

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ABSTRACT

This paper discusses statistical modelling for data with a hierarchical structure, and distinguishes in this context between three different meanings of the term hierarchical model: to account for clustering, to investigate variability and separate predictive equations at different hierarchical levels (multi-level analysis), and in a Bayesian framework to involve multiple layers of data or prior information. Within each of these areas, the paper reviews both past developments and the present state, and offers indications of future directions. In a worked example, previously reported data on piglet lameness are reanalyzed with multi-level methodology for survival analysis, leading to new insights into the data structure and predictor effects. In our view, hierarchical models of all three types discussed have much to offer for data analysis in veterinary epidemiology and other disciplines.

1. Introduction

In everyday language, a hierarchy may be understood as "a system in which people are put at various levels or ranks according to their importance" (Cambridge International Dictionary of English, 1995). In this definition we may replace "people" by other—concrete or abstract—items, and "importance" can also be understood broadly as relating to a certain order in which we view the items, e.g. in taxonomy. Given such a broad definition and usage of the term hierarchy, it is hardly surprising that the term "hierarchical (statistical) model" is used in a variety of different contexts and meanings. Our main focus here is on hierarchical data structures in which the "subjects" (experimental or measurement units) are organized in groups that can be described by, or depicted in, a hierarchy. The organization into groups may follow from the physical location of the subjects or from the circumstances of the recording of data. Typical examples from veterinary studies involve records on animals housed in farms or treated at veterinary clinics or hospitals. Data hierarchies can comprise multiple levels, e.g. by considering sub-units within farms such as pens, or by considering further grouping of farms into regions. Longitudinal data may be seen as a special case of a hierarchical data structure, with repeated measures taken on subjects, where special consideration is needed for potential autocorrelation within subjects over time. In the usual nomenclature, the lowest hierarchical level corresponds to the unit of the observations (or measurements); for example, level 1 may correspond to animals and level 2 to farms. As already indicated, a complex data structure can also be comprised of several unrelated hierarchies pertaining to different characteristics of the subjects; a typical example is that in addition to the location of animals their origin forms another relevant hierarchy. We refer to Dohoo et al. (2009, Chapter 20) for further examples and illustrations of hierarchical and related data structures.

Traditionally the importance of hierarchical data structures has been linked to the violation of independence assumptions involved in classical statistical models, such

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as linear (regression) models. It is intuitively obvious that subjects linked by their presence at the same hierarchical level, e.g. animals in the same farm, may no longer give rise to observations that can be assumed independent. The phrase "animals in the same farm are more similar than animals in different farms" will have made its way through innumerable classrooms. The term clustering is also commonly used to express the notion that similarity between observations sharing certain hierarchical levels leads to clusters in the data. This should not be confused with cluster analysis, which aims at identifying clusters in the data without referring to known structures. Notwithstanding the validity of the assertion (the violation of assumptions), and the need to adjust the modelling approach to account for it, this particular consequence of a hierarchical data structure has to some extent overshadowed the new potentials offered by complex data structures. Modelling and exploring dependence (or correlation) structure may be perceived as more complicated than building predictive equations from a set of predictor variables. It does however offer different kinds of insights into the factors affecting the variability of outcomes of interest across a population. Multi-level modelling utilizes the decomposition of variability across the hierarchical levels to study the impact of predictor variables through separate modelling equations at each level of the hierarchy.

Even after restricting the coverage of hierarchical models to involve modelling derived from particular data structures, we will for the present discussion further distinguish between 3 major domains or scopes of modelling.

(i) Models to account for hierarchical data structure, or clustering (Section 2).
(ii) Simultaneous modelling at multiple hierarchical levels, or multi-level analysis (Section 3).
(iii) Bayesian hierarchical modelling, with multiple layers of equations and assumptions (Section 4).

The objective of the paper is to provide insight into the current state of statistical theory and applications for each of these domains of hierarchical models, with particular focus on their application to veterinary epidemiology. For this purpose we briefly outline historical developments and explain the fundamentals of the modelling, without reproducing any of the detailed expositions from the current literature. Instead, we indicate some recent and ongoing developments by examples from our practice. This paper is based on a presentation given at the 2012 Calvin W. Schwabe Symposium honouring the lifetime achievement in veterinary epidemiology and preventive medicine of Dr. Ian R. Dohoo.

2. Hierarchical models I: to account for clustering

The most commonly used and most versatile method to account for dependence derived from hierarchical data structures consists in including random effects in the statistical model. As a general rule each hierarchical level above the lowest (observation) level should be represented by a set of random effects. Random effects are latent (unobserved) variables with assumed distributions, most commonly Gaussian (normal) distributions, that reflect the variability in the population the random effects represent (e.g. a population of farms). As the random effects are shared by all observations within the same unit of a hierarchical level (e.g., all animals in a given farm), they induce a dependence between such observations. Accordingly, statistical inference based on a random-effects model accounts for clustering in the data.

Random effects in linear models date back at least to work in the 1940s on genetics and animal breeding. A substantial body of statistical theory was developed in the 1950s and 1960s to estimate variance components (the variances of the different random effects), as described in a subsequent review (Robinson, 1991). The assumed Gaussian distributions in linear models with random effects made the calculations manageable with limited computing power, while random-effects models for non-normal data started to appear in theory and applications in the 1980s. This development made use of the formulation of a generalized linear model (GLM) framework, on which the alternative generalized estimating equations (GEE) methodology was also based. Software implementations gradually became available in the 1990s, starting with models involving two hierarchical levels. A seminal paper introducing the new methods for dealing with clustered data to veterinary epidemiology was McDermott et al. (1994), later followed by Dohoo et al. (2001).

The inclusion of random effects in GLMs as well as GEE estimation are by now mainstream approaches, used routinely in veterinary epidemiology. The applied researcher can choose from a multitude of textbooks and implementations in statistical software, although some differences still exist in estimation methods and flexibility. Further developments in recent years have been confined to "difficult situations" (e.g. due to large data sets with many hierarchical levels or to non-standard data structures) and to models beyond the standard GLM framework. We specifically discuss challenges and developments in two commonly encountered areas: time-to-event (also known as "survival") outcomes, and "composite" outcomes that may be viewed as consisting of two (or more) distinct components. Evidently the question of adjusting for hierarchical structure may be asked (and also has been addressed in the literature) for a wide range of analytical settings, from diagnostic-test evaluation and agreement to multivariate analysis.

2.1. Models for composite outcomes

Here we are concerned not with a truly multivariate outcome but with univariate distributions formed by adjoining outcomes of different types. The most common example we have in mind is a quantitative outcome, either a count or a measurement, that can also take values at zero, at a lower detection threshold or at an upper detection threshold. Count distributions such as the Poisson or negative binomial include zero as a legitimate value, but trouble arises if the zeros occur more frequently in the data than predicted from these distributions, a phenomenon broadly referred to as zero-inflation. For continuous outcomes, the presence of measurements equal to a lower detection threshold may be dealt with by considering the distribution as left-censored; for normally distributed non-censored outcomes, these are the so-called Tobit models. Such models may however not fit the data well if the proportion of censored values cannot be predicted reasonably well by the normal distribution estimated from the non-censored data. These examples motivate a modelling approach where the data are thought of as consisting of two components: a zero (lower threshold) component and a non-zero component, and where these two parts are modelled separately.

Let us describe this scenario using general concepts and terminology. A mixture distribution is the combined distribution of heterogeneous outcomes across several subpopulations, typically when there is no (definitive) information about subpopulation membership. A *zero-inflated* Poisson, or negative binomial, distribution/model is a mixture because the zero counts originate from two subpopulations: observations that are systematically zero, and observations that happen to be zero but could also have provided a positive count. For observations of fecal egg counts in parasitology, these two subpopulations could be the animals that are not infected and infected, respectively, with the parasite. The alternative *hurdle* model, which simply separates the counts into zeros and non-zeros and models each part separately, is not a mixture model. The term *zero-altered* model has been used for the zero-inflated and hurdle models (Hilbe, 2011; Atkins et al., 2012). If the data in the non-zero part are of a different type that does not allow for zero values, in particular measurements on an interval scale, we can model the two parts separately. This construction is similar to a hurdle model and does not lead to a mixture model; such models are more generally termed *two-part* models (Olsen and Schafer, 2001). In all these models, the censored (i.e., zero) part is modelled through a logistic regression, whereas the non-censored part can be modelled in a linear or generalized linear framework appropriate for the data type.

Because the two components in a two-part model are modelled separately, they can have different predictors and may also include separate random effects to account for a hierarchical data structure. For the sake of our discussion of such models, let us consider a specific example: data with continuous measurements (Y) of sea lice abundance at salmon aquaculture farms measured weekly over several years (Kristoffersen et al., 2013). The two parts of the model can be represented as,

$$\text{Part I}: \quad \text{logit}(Pr(Y=0)) = X^I\beta^I + u^I_{farm}, \tag{1}$$

$$\text{Part II}: \quad \text{given } Y>0: \quad Y = X^I\beta^{II} + u^{II}_{\text{farm}} + \epsilon,$$

where the errors of the non-censored values (ϵ) could e.g. be drawn from normal, log-normal or gamma distributions, as appropriate, and the farm random effects would usually be taken as normally distributed: $(u^I, u^{II}) \sim \mathrm{N}(0, 0, \sigma_I^2, \sigma_{II}^2, \rho\sigma_I\sigma_{II})$. Because the two modelling scales are entirely different, we would generally expect to have a pair of distinct but correlated random effects for each farm. The correlation (ρ) may be quite strong—in (1) a negative correlation because the two equations are in opposite directions (low and high values, respectively)—and estimation can be difficult if the correlation parameter gets numerically close to 1. Maximum likelihood estimation in models such as (1) can be carried out using general optimization routines and numerical integration for the random effects (e.g., Liu et al., 2010). Software implementations for estimation in two-part random-effects models (or the count data equivalents) are starting to appear but are still somewhat scattered across different programs and packages (Atkins et al., 2012), and do not yet include models with multiple (>2) hierarchical levels. Alternative estimation approaches, such as moment-based or Bayesian estimation (Hasan et al., 2009; Zuur et al., 2012), may be the best or only choices for complex data structures. Besides the access to capable estimation algorithms, other issues still being studied are choice of model or error distribution for the non-censored data (Liu et al., 2010), dealing with autocorrelation in longitudinal data (Zuur et al., 2012), proper interpretation of coefficients and variances in the two equations, and model validation.

2.2. Survival analysis

Early developments in modelling of time-to-event data to account for hierarchical structure date back to the 1980s and early 1990s (e.g. Clayton, 1991; McGilchrist and Aisbett, 1991), and implementations in statistical software enabling analysis of data with two or multiple hierarchical levels appeared around year 2000 (e.g. Ripatti and Palmgren, 2000). Nevertheless, models involving random effects (or *frailties*, their multiplicative counterparts) are still not as easily accessible as for GLMs, as can be gleaned from fairly recent textbook treatments of frailty models (Duchateau and Janssen, 2008; Wienke, 2011). Without attempting a full discussion, a few pointers can be offered for explanation. Survival analysis is a more diverse field than GLMs, with many different models and estimation approaches developed for different situations. The Cox proportional hazards model is perhaps the closest analogue of a GLM, but its (partial) likelihood function does not analytically lend itself to the same ways of integrating random effects, and Gaussian random effects do not have the same computational advantages as in GLMs. Frailties affect the interpretation of regression (fixed-effects) parameters more profoundly than in GLMs. Finally, the presence of time-varying effects or predictors create more complex predictive patterns than in GLMs, even for the Cox model.

We will restrict the present discussion to the Cox model and an estimation approach that reformulates the Cox model as a suitable Poisson regression after splitting the event-time range into the smallest intervals between events, sometimes referred to as the counting process data setup. This approach is justified by an equivalence between the Cox and Poisson regression likelihood functions when no frailties are present, and the Poisson model can be extended to account for hierarchical structure (Ma et al., 2003; Rabe-Hesketh and Skrondal, 2012). Such Poisson models have e.g. been used for modelling time until clinical mastitis among cows in multiple herds (Schukken et al., 2010). One advantage of the approach is its flexibility inherited from the Poisson modelling framework, and it has fared well in simulation studies (Feng et al., 2009). On the downside, the counting process data set may become huge and therefore difficult to manage computationally with standard implementations for random-effects GLMs. However, if the Cox model involves time-varying effects, perhaps as a result of violations of the proportional hazards assumption, the expansion to a counting process format may be inevitable after all, and should not be counted as a drawback of the Poisson approach. Indeed, many of the software implementations for other estimation approaches in frailty Cox models do not allow for the counting process data format and therefore seem of limited use for data exhibiting non-proportional hazards in a Cox model.

As computational power steadily increases, the Poisson modelling approach becomes feasible even when the counting process format data comprise many megabytes, e.g. clinical mastitis data collected from multiple lactations of about 8000 cows in 69 herds across Canada (Elghafghuf et al., 2012). There is also a potential for modelling more complex, non-hierarchical data structures. For example, beef-cattle farms from Western Canada in a large-scale study on animal-health effects associated with exposure to emissions from oil and natural gas field facilities (Waldner, 2008) were located in distinct ecological regions and were serviced by veterinary clinics. Because the veterinary clinics had clients in several regions and some herds were serviced by multiple clinics, the data structure involved a cross-classification between regions and clinics as well as a multiple membership classification in clinics (Browne et al., 2001). Also in many other areas of survival analysis, developments of the last decade have addressed inclusion of hierarchical structure, such as interval censored observations (Sun, 2006), populations with a fraction of cured subjects (Peng and Zhang, 2008; Rondeau et al., 2011), and dynamic models for time-varying covariates (Martinussen and Scheike, 2006).

3. Hierarchical models II: multi-level analysis

As defined previously, multi-level modelling and analysis involves equations at multiple levels of a hierarchy. The idea that relationships may exist at different levels dates back long in sociology, involving terms such as group-level vs. individual-level regression, ecological correlation (Robinson, 1950) and contextual analysis. The suggestion to combine regressions at different levels into a single mixed model and analysis was put forward by several authors in the mid-1980s, and soon after the first versions of software purposively developed for this new multi-level analysis started to appear. This development eventually crystallized into two major software packages developed by different research groups linked to applications in education and sociology: MLn (later MLwiN; Goldstein, 1987) and HLM (Bryk and Raudenbush, 1992). Historically we may see multi-level analysis as "the marriage of contextual analysis and traditional statistical mixed model theory" (de Leeuw and Meijer, 2007). The potential of multi-level analysis for epidemiological studies was discussed around the turn of the century (Greenland, 2000).

Examples of multi-level analyses can be found in veterinary epidemiology, but with no tradition for displaying models by means of multiple equations, this type of analysis may be difficult to identify. The distinguishing features of such multi-level models are random slopes (e.g. Amezcua et al., 2010), or contextual effects (explored in e.g. Mörk et al., 2009; Kujala et al., 2010), or both. To elucidate the characteristics of multi-level models, let us, within a farm–animal hierarchy, outline both the combined, and the farm- and animal-specific regression equations. For simplicity, we consider a linear mixed model for a continuous outcome Y and two additive, quantitative predictors, one at the animal level (x_1) and one the farm level (x_2). The simplest model (termed a "random-intercept" model, but in our usage a method to account for clustering rather than a genuine multi-level model) includes only farm random effects (u_j for farm j) in addition to the fixed-effects predictor for animal i at farm j,

$$Y_{ij} = \beta_0 + \beta_1 x_{1ij} + \beta_2 x_{2j} + u_j + \epsilon_{ij}, \tag{2}$$

with the commonly used assumptions: $u_j \sim N(0, \sigma_u^2)$ and $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$. In (2), the slope for x_1 is constant across farms; a *random slope* (or random coefficient) for the regression on x_1 is an additional term $b_{1j} x_{1ij}$ where the random fluctuation (b_{1j}) of the slope at farm j may be assumed drawn from $N(0, \sigma_b^2)$. A *contextual* effect for x_1 is an additional slope (say β_3) for the regression on the farm averages ($\bar{x}_{1\cdot j}$), the implications of which will be explained below. With these two extra terms added to (2), the equation takes the form

$$Y_{ij} = \beta_0 + \beta_1 x_{1ij} + b_{1j} x_{1ij} + \beta_2 x_{2j} + \beta_3 \bar{x}_{1\cdot j} + u_j + \epsilon_{ij}. \tag{3}$$

We see that the resulting slope for x_1 in (3) is $\beta_1 + b_{1j}$, and by taking this collection of terms a bit further we arrive at a representation of the model (3) in separate equations at the animal and farm levels:

$$\begin{aligned} Y_{ij} &= \beta_{0j} + \beta_{1j} x_{1ij} + \epsilon_{ij}, \quad \text{where} \\ \beta_{0j} &= \beta_0 + \beta_2 x_{2j} + \beta_3 \bar{x}_{1\cdot j} + u_j, \quad \text{and} \quad \beta_{1j} = \beta_1 + b_{1j}. \end{aligned} \tag{4}$$

The two equations of (4) specify the modelling of the intercept and slope of the first equation, both depending on farms. Note that the farm-level predictor appears only in the farm-level part. The animal-level predictor, however, appears in both equations because of the contextual effect that allows for different (average) slopes of x_1 at the animal and farm levels. We can explicitly represent the farm-level regression by working out the resulting equation from (3) for the farm means ($\bar{Y}_{\cdot j}$),

$$\bar{Y}_{\cdot j} = \beta_0 + \beta_1 \bar{x}_{1\cdot j} + \beta_2 x_{2j} + \beta_3 \bar{x}_{1\cdot j} + u_j + \bar{\epsilon}_{\cdot j}. \tag{5}$$

Hence, the farm-level slope in (5) for x_1 is ($\beta_1 + \beta_3$), and it equals the within-farm average slope (β_1) only if the contextual term cancels (i.e., $\beta_3 = 0$). All these equations are of interest only if predictors actually have different effects at different levels, but in some research areas (e.g. education) this has been stated as the rule rather than the exception (Snijders and Bosker, 1999, p. 56). In such instances, representing the model by equations at separate levels may be intuitively more appealing than the single combined equation (3).

The methodology for analysis of multi-level linear and generalized models is fully developed and available both in specialized multi-level software and with most implementations of random-effects models in general-purpose statistical software. The relative sparsity of applications of multi-level analysis in veterinary epidemiology can therefore be attributed either to lack of interesting applications or to unfamiliarity with these models and their proper interpretation. The random-intercept model (2) to account for clustering is much more frequently encountered in the literature than multi-level analysis, and is indeed sometimes reported without mention of the variance parameters. In our view, better exploration and presentation of the information inherent in data with a hierarchical structure should be encouraged. In parallel with the extension of random-effects models beyond the GLM framework, as discussed in Section 2, we may also see applications of multi-level analysis in these new settings. We next reanalyze data from a previous published study to provide an example of random slopes and contextual effects in survival analysis.

3.1. Example: multi-level survival analysis of lameness in piglets

A study on lameness in piglets was carried out during 5 months in 1990–91, and first described by Christensen (1996). The study comprised 7632 litters of 5436 sows in 35 swine farms in Denmark; we consider here only one (i.e., the first) litter per sow. Lameness among the piglets in a litter was monitored by producers; and the time of the first treatment with antibiotics of piglets in a litter for lameness was recorded, and serves as the "survival time" for the litter. A total of 678 litters (12.5%) were treated. The event times ranged from 1 to 38 days with a median of 10 days. The litters were generally followed until weaning, no later than 40 days after farrowing. Predictors of interest were: farm size (3 categories), farm-health status (0/1; 1 = high health status, such as SPF (specific-pathogen-free)), parity (with parities ≥8 grouped into one category), litter size, sow treatment (0/1; producer treatment of sows prior to or around farrowing), and floor area per sow. Christensen (1996) reported a preliminary analysis by an ordinary Cox model (without adjusting for the hierarchical structure), in addition to a logistic regression analysis for lameness that included farm random effects. A subsequent study (Josiassen and Christensen, 1999) further investigated the data by different methods to account for farm clustering in the logistic model, and confirmed the initial finding that the sow treatment changed from being (significantly) a risk factor to being protective when the clustering was adjusted for properly. The purpose of the present analysis is to investigate the effects of adjusting for clustering in the Cox model and to discuss the results of a full multi-level survival analysis, with special focus on the sow-treatment predictor. For further details about the data we refer to the previously published papers.

All predictors were analyzed in both ordinary Cox models, including assessment of the proportional hazards assumption based on the scaled Schoenfeld residuals (Dohoo et al., 2009), and in Cox models with farm random effects (shared frailty models). For the latter, we used the penalized partial likelihood estimation approach implemented in Stata (version 12; www.stata.com). Among the fixed effects, only the farm-health status showed non-proportional hazards; after the predictive equation was augmented with its interaction term with ln(time), no indication of non-proportionality remained. The only significant litter-level predictor was sow treatment. Multi-level Cox models with both farm-level random slopes and a contextual term for sow treatment were analyzed using the Poisson estimation approach (Section 2.2), whereby estimation in the Poisson model used Gaussian quadrature for the random effects.

Although only discussed sporadically in the literature (e.g., Yau and McGilchrist, 1998; Manda and Meyer, 2005), shared frailty models imply an additional assumption of proportional hazards for the random-effect terms; in our example, the (random) farm differences must remain constant over time. To our knowledge, no general procedures or tests to assess this assumption exist, but the Poisson modelling framework allows us to estimate interactions between the farm random effects and functions of time. Such interaction terms are also random effects and hence play a similar role in the model as random slopes. In practice, we usually limit the exploration of time-dependent effects to original and log-transformed time scales and select the one with the best fit; in our example, random-effect interactions with ln(time) were included in the model. Summing up, all models fit can be described by the following equation for the hazard $h_{ij}(t)$ of litter i in farm j at time t, where x_1 = sow treatment and x_2 = farm-health status,

$$\ln h_{ij}(t) = \ln h_0(t) + \beta_1 x_{1ij} + b_{1j} x_{1ij} + \beta_2 x_{2j} + \beta_3 \bar{x}_{1\cdot j} + \beta_4 x_{2j} \ln(t) + u_j + c_j ln(t), \tag{6}$$

with the random-effect assumptions: $(u_j, b_{1j}, c_j) \sim N(0, \Sigma)$ where the variance-covariance matrix Σ contains the 3 random-effect variances $(\sigma_u^2, \sigma_b^2, \sigma_c^2)$ in the diagonal and the pairwise covariances $(\sigma_{ub}, \sigma_{uc}, \sigma_{bc})$ outside the diagonal. For computational efficiency, $\ln(t)$ was centred by subtracting its approximate value in the split data (i.e. 2.4, corresponding to $t \approx 11$ days). To avoid introduction of additional notation, we abstain from displaying the actual Poisson modelling equation, and the results below exclude the regression coefficients for our fourth-order polynomial of time to approximate the baseline hazard h_0 (Rabe-Hesketh and Skrondal, 2012). Parameter estimates (with standard errors) for 6 different models obtained from (6) by restricting some terms or parameters to zero are shown in Table 1. For comparison of the different models, the table also includes their optimal log-likelihood values. We discuss different aspects of the results under separate subheadings. All interpretations of parameter estimates are based on the full model (6).

3.1.1. Accounting for hierarchical structure

The within-farm clustering was strong, as witnessed by the numerically large farm variance and the vast improvement in fit ($\ln(L)$) by the random effects. The inference for the farm-level predictor (health status) was substantially affected by the random effects, showing both clustering and confounding effects, although the statistical

Table 1
Estimates with standard errors (SE) for analyses of Danish piglet lameness data by Cox proportional hazards models obtained as subsets of the full multi-level model.

Parameter/statistic	Ordinary Cox model	Farm random effects		Multi-level sow tx effect		Full model
		Prop. haz.	Time-dep.	Random slope	Contextual	Eq. (6)
β_1: sow tx	0.21 (.08)	−0.16 (.09)	−0.16 (.09)	−0.63 (.24)	−0.18 (.09)	−0.27 (.16)
β_3: context.	n/a	n/a	n/a	n/a	3.68 (.87)	4.35 (.93)
β_2: hstatus	0.28 (.08)	0.73 (.48)	0.82 (.50)	0.63 (.43)	0.77 (.46)	0.81 (.47)
β_4: $\times \ln(t)$[a]	−0.36 (.11)	−0.37 (.11)	−0.69 (.32)	−0.62 (.30)	−0.80 (.32)	−0.85 (.32)
σ_u^2: farms	n/a	1.72 (.51)	1.79 (.54)	1.58 (.49)	1.49 (.47)	1.60 (.45)
σ_c^2: $\times \ln(t)$[a]	n/a	n/a	0.50 (.20)	0.49 (.19)	0.52 (.21)	0.53 (.21)
σ_{uc}: covar.	n/a	n/a	−0.50 (.29)	−0.44 (.27)	−0.72 (.29)	−0.71 (.29)
σ_b^2: r. slope	n/a	n/a	n/a	0.22 (.21)	n/a	0.17 (.14)
σ_{ub}: covar.	n/a	n/a	n/a	0.59 (.30)	n/a	−0.10 (.24)
σ_{bc}: covar.	n/a	n/a	n/a	−0.18 (.13)	n/a	−0.11 (.12)
$-\ln(L)$[b]	4188.75	3935.30	3896.24	3892.04	3889.65	3886.38

[a] Centred at the average value $\ln(t) = 2.4$, i.e. $t \approx 11$ days.
[b] Log-likelihood value from Poisson regression model.

significance remained. The sow-treatment effect shifted from significant risk to protection—the same pattern as discussed above for the logistic analysis.

3.1.2. Non-proportional hazards for farm effects

Simple descriptive procedures for assessing farm effects over time include tests and graphs with farms as a (fixed) categorical predictor. To illustrate, Fig. 1(a) gives Kaplan–Meier survival curves for a sample of 5 of the 35 farms (all with low farm-health status and including the farm with the lowest overall survival) and demonstrates substantial variation between farms in both their overall risk and in the times at which the largest drops in survival probabilities occur. Hence based on this graph, we could anticipate the time-proportionality between farms to be violated. A Nelson–Aalen plot of the cumulative hazards, preferably on log-scale, is visually more effective to assess the proportionality assumption (Dohoo et al., 2009) but the Kaplan–Meier plot is easier to interpret and gives additional useful information about the survival across farms.

The estimates in Table 1 testify to substantial variability in the coefficients for $\ln(t)$ across farms and a strong improvement in model fit by the time-dependent farm effects. As the modeling of time includes also the $\ln(t)$ and polynomial terms for the baseline hazard, it is difficult to interpret the variance, σ_c^2, directly. The covariance, σ_{uc}, between an intercept and a slope for a centred predictor would often be expected to be close to zero, but correlations were here moderate and negative (e.g. $-0.71/\sqrt{1.60 \cdot 0.53} = -0.77$). Fig. 1(b) shows the predicted log-hazards for non-treated sows in the same 5 farms; these log-hazard curves are seen to be clearly non-parallel among the farms (i.e., showing evidence of the time-dependence and the non-proportionality captured by the modeling).

3.1.3. Effects of farm-health status on lameness

The estimated effects for farm-health status (β_2) and its time-dependent component (β_4) were fairly constant across all models involving farm random effects, even if the addition of time-dependent farm effects inflated the estimates for the latter. The estimated hazard for lameness was significantly higher early after farrowing in farms with high rather than low health status (e.g. the hazard ratio (HR) at 5 days equalled $HR = e^{0.81-0.85(\ln(5)-2.4)} = e^{1.48} = 4.40$) and declined over time (e.g. $HR = e^{0.81} = 2.25$ at 11 days) until equal hazards were reached towards the end of the follow-up period (29 days).

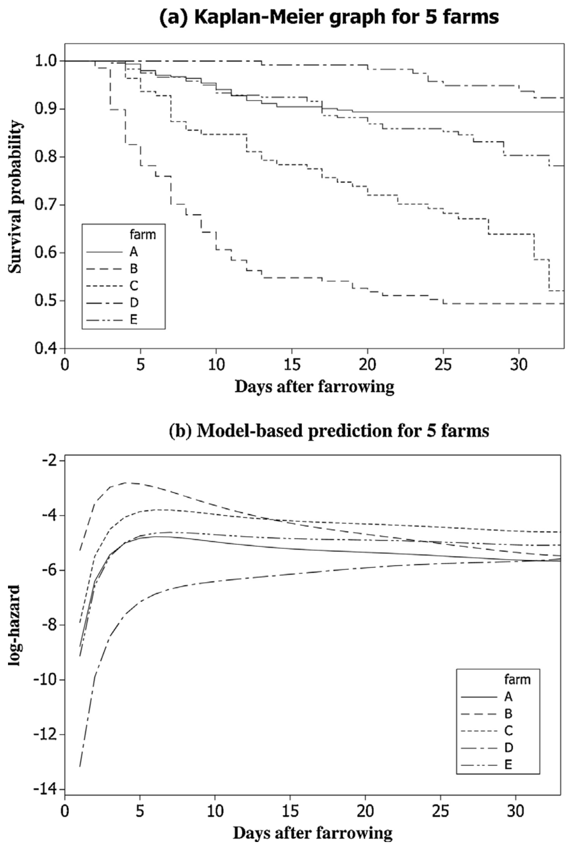

Fig. 1. For 5 selected Danish swine farms (with low farm-health status), (a) non-parametric estimated survival (until litter lameness), and (b) estimated log-hazard for untreated sows, based on the full multivariable, random effects Cox model (Eq. (6)).

3.1.4. Random slope for sow treatment

The addition of random slopes to the shared frailty model led to relatively small, but statistically significant (by likelihood-ratio tests) improvements in model fit. The random slopes substantially affected the sow-treatment parameter (β_1), although counterbalanced by the further addition of a contextual effect (below). The average sow-treatment effect was slightly protective but weak ($P = 0.10$). An estimated 95% range for the random slope (i.e., the sow-treatment effect) in a specific farm could be computed as $-0.27 \pm 1.96\sqrt{0.17} = -0.27 \pm 0.81$, hence spanning both risk and protective effects across the farms. In such a situation, further investigation into which farms were associated with risk versus protection is warranted. For a binary predictor such as sow treatment, there is usually no specific expectation as to the strength and direction of the association between intercept and slope, but nevertheless one would usually allow for a covariance between them. After the contextual effect was incorporated, the estimated correlation was weak ($-0.10/\sqrt{1.60 \cdot 0.17} = -0.19$).

3.1.5. Contextual effect for sow treatment

The farm averages for the binary sow-treatment predictor, the proportions of sows treated at each farm, varied strongly between farms (range 4–85%). The addition of a contextual effect for sow treatment resulted in a larger improvement in model fit than the random slopes, and the corresponding estimates were numerically large. This would indicate that the farm proportion of treated sows was a useful predictor to distinguish between farms with low and high hazards, and accordingly the (co)variance parameters associated with sow treatment (i.e., $\sigma_b^2, \sigma_u^2, \sigma_{ub}$) decreased somewhat in the presence of the contextual effect. The estimated contextual effect implied that a (modest) 10% difference in proportion of sow treatments was associated with a hazard ratio of $HR = e^{0.1 \cdot 4.35} = e^{0.435} = 1.54$. In other words, farms with high proportions of treated sows also had higher hazards of lameness. Together with our previous discussion of the within-farm effects of sow treatment, these estimates illustrate the different effects of sow treatment at animal and farm levels.

3.1.6. Biological interpretation of the findings for "lameness"

At the time of the study, lameness in the preweaning period was predominantly caused by joint infections, and "the most important factors determining the prevalence of lameness could be the presence of infectious agents and management of infectious diseases in the herd" (Christensen, 1996). Both presence and management would be latent factors that were part of the random farm effect in the previously reported (logistic) analyses. The present analysis helps us to unravel this farm effect.

First, we identified that the farm effects varied over time (time-varying farm effects). The most likely explanation is that "treatment of lameness by producers" is a multifactorial disease (or condition), and a necessary cause may be reached in a different manner in each farm. For example, the two farms B and C with approximately the same overall proportions of litters treated for lameness differed in the onset of lameness. In the first farm, about 45% of the litters were treated for lameness before day 14, as would be the situation if neonatal joint infection was a serious and dominating health problem, e.g. when a disease of infectious nature first is spreading in a fully susceptible population. The other farm had almost the same overall proportion of litters treated for lameness, but the onset of disease was gradual. Therefore the cause of lameness may have been more complex and have included factors such as gradual decline in maternal antibodies in chronically infected farms, slow transmission of infection, and environment factors (e.g. abrasions from slatted floors).

Second, the new analysis confirmed the earlier impression that SPF (high health status) farms had a higher hazard of treatment for lameness than conventional (low health status) farms; however, that effect was time-dependent. The obvious explanation would be that an agent causing joint infections was circulating in the SPF farms (only) at the time. However, we caution and hypothesize it might be a reporting bias rather than a truly higher incidence of lameness in SPF farms. The producer could only record one clinical sign as the cause of treatment, but if an infectious agent causing lameness in piglets circulated in all farms, then the lameness could have been masked by occurrence of other diseases. For example, if one or two piglets in a litter had a joint infection and 8 had diarrhea, the producer would treat the whole litter for diarrhea and record the treatment as diarrhea. At the time of the study, diarrhea and mortality were perceived as the important diseases before weaning, and producers may have been more diligent in recording them, thereby masking the occurrence of lameness. Preweaning mortality was at its highest in the first 3 days of life and diarrhea in the first week, and both were more prevalent in conventional farms. Therefore they might have masked a higher incidence of lameness early in the preweaning period in conventional farms.

Third, the new analysis added to our understanding of the sow-treatment effect, specifically how it was linked with the farm effect. Some farms clearly had a high proportion of treated sows and a high hazard of lameness. Whether that was a result of differences in presence of infectious agents in the farms or of different treatment thresholds remains unclear in retrospect, but would have warranted further investigation. We can now control this farm effect on sow treatment and assess the within-farm effect of sow treatment. The modest protective effect of sow treatment on lameness in the litter, with a substantial variation between farms, adds to our understanding. The producer may have treated a healthy or a sick sow, and the treatment may have been effective or not. The true health status of the sow was a latent entity that may have depended on the producer's ability to recognize and treat health problems in sows and on the presence of infectious agents in the farm. This could explain the variation in the sow-treatment effect between farms. When the farm effect was controlled, the hazard for lameness was lower in litters of treated sows than untreated sows. The treated sows may have had a better health status, potentially affecting the lameness in the litters in two ways: healthy sows may have been better able to care for the litter, and they may have been less likely to transmit joint infections to the litter.

In summary, the data offered a (possibly first) illustration in veterinary epidemiology of random slopes and contextual effects as well as time-dependent farm effects for a survival outcome. The multi-level analysis in our view offered valuable insights into the structure and relationships in the data beyond the simple designation of sow treatment as a protective or risk factor.

4. Hierarchical models III: Bayesian context

Bayesian statistics has undergone a fascinating historical development from its inception in the 18th century over its fall into (partial) obscurity in the 20th century to its surge in the last two decades (Fienberg, 2006). The recent ever-growing popularity of Bayesian statistics has been driven by the advent of simulation-based estimation procedures, first and foremost Markov chain Monte Carlo (MCMC) algorithms, and by steady advances in computing power. Early efforts were made to make the new MCMC methodology widely accessible (e.g. in the free BUGS and WinBUGS software). In recent years Bayesian methods have become mainstream to a much wider audience, due both to the publication of a wealth of new textbooks with a more applied focus (e.g. Christensen et al., 2011; Lunn et al., 2012) and to the incorporation of Bayesian estimation algorithms into standard statistical software. Bayesian statistics has in the past decade delivered the foundation of diagnostic-test evaluation in the absence of a gold standard, a development largely driven by the research group on Bayesian Epidemiologic Screening Techniques (www.epi.ucdavis.edu/diagnostictests/) at the University of California, Davis (e.g. Branscum et al., 2005). Bayesian statistics has also been applied to many other areas of veterinary epidemiology, typically for problems that could not be easily addressed with classical (frequentist) statistical methods, for example in spatial statistics. It seems likely that the Bayesian approach will gain popularity also for "simple" problems, as these methods become commonly taught in biostatistics and epidemiology.

The term hierarchical has been used in the Bayesian framework to denote models specified in multiple layers. Recall that any Bayesian model involves an observation model (determining the likelihood function, given the parameters) and a prior distribution (or model) for the parameters. As described by Christensen et al. (2011, Section 4.12), the hierarchical nature of a model may manifest itself in either of these two parts of the model. If the observation model involves latent variables, such as random effects corresponding to a hierarchical data structure, the model assumptions can be formulated through: (i) distributions conditional on the latent variables, and (ii) distributions for the latent variables. A random-effects logistic regression model is usually formulated this way. All the models related to hierarchical structure discussed in Sections 2 and 3 hence become Bayesian hierarchical models if analyzed in a Bayesian framework. Latent variables obviously occur in many other contexts, making the scope of Bayesian hierarchical models much broader than just dealing with hierarchical data structures. Multiple layers occur in the prior distribution if it involves parameters which in turn are drawn from another distribution. The terms hyperparameters and hyperpriors are commonly used for this construct, whose primary function is to facilitate the specification of the prior distribution. These two ways of thinking about hierarchical Bayesian models have considerable overlap. As a classical example, if we model counts of events in groups (say herds) of size n_i by binomial distributions Bin(n_i, p_i), we might think of the herd proportions as latent variables, and we could think of the parameters in their distribution, say a Beta distribution (α, β), as the hyperparameters with a suitable hyperprior distribution (Gelman et al., 2003). A multi-layered model specification is particularly attractive when using the Bayesian framework to model large and/or complex systems (Carlin et al., 2006). The latter authors present hierarchical Bayesian models in a generic form whereby the distribution (represented by $f(\cdot)$) of the data, the process (behind the data), and the parameters is written as the product of the three terms: f(data|process, parameters), f(process|parameters), and f(parameters). Each of these terms may be specified in multiple layers as well. The use of a Bayesian framework for quantative risk assessment has been described in a couple of recent monographs (Kelly and Smith, 2011; Fenton and Neil, 2012).

We conclude this brief review of Bayesian hierarchical modelling with a reference to a fairly recently developed Bayesian estimation method that *does not* rely on MCMC estimation, employing instead a so-called integrated nested Laplace approximation (INLA) procedure (Rue et al., 2009). This analytical framework includes normally distributed latent variables, and thus allows for hierarchical data structure, but is targeted towards complex applications involving temporal and spatial smoothing where MCMC estimation may be too difficult to apply (requiring specialized MCMC samplers) or prohibitively slow. An implementation of the procedures has been made available as a comprehensive library for the R statistical software, with an extensive supporting website (http://www.r-inla.org). Applications of the INLA methodology in veterinary epidemiology have been published (e.g. Schrödle et al., 2011), and more are certain to follow.

5. Conclusions

Hierarchical models (in any of the three senses discussed) are of paramount importance for data analysis in veterinary epidemiology, in part due to the hierarchically structured food animal production systems. These models have the potential to develop further as new computational techniques and resources become available, and will likely play an important role as solutions are sought to the emerging demand for analysis of complex systems. Hierarchical models are well integrated into veterinary epidemiological research, thanks to pioneers both in their use and in dissemination of their utility. Among these, we regard the 2012 Calwin W. Schwabe Award recipient, Dr. Ian R. Dohoo.

Conflict of interest statement

The authors have no conflict of interest related to this work.

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